

Logical Coherence Score: Experimental Evaluation

FactReasoner — LCS experiments

July 14, 2026

1 Setup

This report evaluates the Logical Coherence Score (LCS) pipeline over 9 worked examples from `data/lcs`, across 3 model(s) (llama-4-maverick-17b, llama-3.3-70b-instruct, gpt-oss-120b), 3 conditional-strength uncertainty-quantification method(s) (surr-lp, surr-smp, verbal). Relations are mined with the **all** candidate-pair policy. Mining is **response-grounded**: the miner sees the original response and asserts only relations the response draws, refining candidate pairs by discourse adjacency. For every mined coherence MRF all four LCS readouts are computed: mean-marginal, consistency, reified, log-partition.

2 Dataset

Table 1: Per-example size: atoms and mined relations. The **Code** column is the short label used in the per-score tables.

Example	Code	Atoms	Relations
aeroparts	aero	16	57 (3.6)
ex1-damages	e1-dmg	13	38 (2.9)
ex2-biography	e2-bio	19	46 (2.4)
ex2-biography-contradicted	e2-con	12	12 (1.0)
ex3-narrative	e3-nar	33	89 (2.7)
ex4-summary	e4-sum	15	68 (4.5)
ex5-renda-K	e5-K	15	68 (4.5)
ex5-renda-S	e5-S	18	59 (3.3)
ex6-incident	ex6-incident	13	66 (5.1)

3 Results

4 Findings

Averaged over all model/example cells, the headline mean-marginal LCS by conditional-strength method is: surr-lp=0.616, surr-smp=0.512, verbal=0.787. Differences here reflect how each UQ method sets the edge strengths that drive the coherence MRF; the verbalized baseline is included for comparison.

Contradiction-heavy examples average mean-marginal 0.508 versus 0.676 for the cleaner ones, consistent with the LCS being pulled down when the response asserts a live internal conflict.

Table 2: LCS score **mean_marginal**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

Model	Config	aero	e1-dmg	e2-bio	e2-con	e3-nar	e4-sum	e5-K	e5-S	ex6-incident
llama-4-maverick-17b	surr-lp	0.56	0.69	0.92	0.37	0.37	0.13	0.11	0.20	0.83
	surr-smp	0.12	0.33	0.94	0.55	0.33	0.03	0.00	0.25	0.35
	verbal	0.81	0.97	0.92	0.43	0.76	0.76	0.91	0.81	1.00
llama-3.3-70b-instruct	surr-lp	0.84	1.00	0.99	0.50	0.58	0.11	0.89	0.22	0.88
	surr-smp	0.11	1.00	1.00	0.00	0.12	0.20	0.10	0.17	0.04
	verbal	0.81	0.66	0.98	0.49	0.61	0.50	0.72	0.70	0.96
gpt-oss-120b	surr-lp	0.90	0.99	0.91	0.56	0.92	0.22	0.05	0.99	0.88
	surr-smp	0.86	1.00	0.91	0.65	0.92	0.88	1.00	0.98	1.00
	verbal	0.79	0.99	0.82	0.44	0.87	0.77	0.94	0.81	1.00

Table 3: LCS score **consistency**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

Model	Config	aero	e1-dmg	e2-bio	e2-con	e3-nar	e4-sum	e5-K	e5-S	ex6-incident
llama-4-maverick-17b	surr-lp	0.15	0.00	1.00	0.17	1.00	1.00	1.00	0.99	1.00
	surr-smp	1.00	1.00	1.00	0.50	1.00	1.00	1.00	0.69	1.00
	verbal	0.83	0.00	1.00	0.93	0.36	0.82	0.11	0.04	1.00
llama-3.3-70b-instruct	surr-lp	0.00	0.00	1.00	0.42	1.00	0.34	0.00	1.00	0.00
	surr-smp	1.00	0.00	1.00	0.43	1.00	0.45	1.00	1.00	1.00
	verbal	0.20	0.66	1.00	0.42	0.45	0.34	0.12	0.03	0.00
gpt-oss-120b	surr-lp	0.00	0.00	1.00	0.20	0.00	1.00	0.75	0.00	0.05
	surr-smp	0.00	0.00	1.00	0.02	0.00	0.00	0.00	0.00	0.00
	verbal	0.01	0.00	1.00	0.86	0.00	0.32	0.00	0.00	0.00

Table 4: LCS score **reified**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

Model	Config	aero	e1-dmg	e2-bio	e2-con	e3-nar	e4-sum	e5-K	e5-S	ex6-incident
llama-4-maverick-17b	surr-lp	0.13	0.15	0.50	0.50	0.43	0.50	0.46	0.50	0.02
	surr-smp	0.50	0.50	0.50	0.50	0.39	0.50	0.50	0.43	0.50
	verbal	0.20	0.01	0.43	0.42	0.09	0.25	0.01	0.00	0.49
llama-3.3-70b-instruct	surr-lp	0.40	0.31	0.50	0.06	0.36	0.44	0.00	0.50	0.50
	surr-smp	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.42	0.50
	verbal	0.18	0.28	0.49	0.06	0.16	0.01	0.00	0.00	0.02
gpt-oss-120b	surr-lp	0.50	0.50	0.50	0.50	0.50	0.50	0.49	0.00	0.05
	surr-smp	0.43	0.50	0.50	0.30	0.47	0.27	0.38	0.16	0.00
	verbal	0.02	0.06	0.33	0.44	0.12	0.03	0.04	0.02	0.05

Table 5: LCS score **log_partition**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

Model	Config	aero	e1-dmg	e2-bio	e2-con	e3-nar	e4-sum	e5-K	e5-S	ex6-incident
llama-4-maverick-17b	surr-lp	0.15	0.00	1.00	0.31	0.69	0.09	0.19	0.07	1.00
	surr-smp	0.30	0.03	1.00	0.59	0.67	0.03	0.00	0.13	1.00
	verbal	0.35	0.02	1.00	0.57	0.57	0.09	0.09	0.18	1.00
llama-3.3-70b-instruct	surr-lp	0.16	0.01	1.00	0.00	0.49	0.01	0.22	0.03	1.00
	surr-smp	0.06	0.02	1.00	0.00	0.11	0.00	0.33	0.08	0.50
	verbal	0.22	0.17	1.00	0.00	0.34	0.02	0.16	0.10	0.14
gpt-oss-120b	surr-lp	0.23	0.09	1.00	0.49	0.80	1.00	0.01	0.00	0.51
	surr-smp	0.49	0.12	1.00	0.45	0.87	0.68	0.00	0.14	0.00
	verbal	0.31	0.01	1.00	0.39	0.66	0.62	0.10	0.16	0.00

Across examples the four readouts differ in dynamic range mean_marginal (spread 1.00), consistency (spread 1.00), reified (spread 0.50), log_partition (spread 1.00); **mean_marginal** is the most discriminative in this run.

5 Case study: an authored coherent response

Design. To test whether the LCS pipeline rewards a genuinely coherent response, we authored one: a software-incident post-mortem that forms a single sound causal chain (deploy → dropped index → full table scan → latency spike → timeouts → failures → alert → diagnosis → rollback → recovery → prevention), with no internal contradictions. It was run through the same matrix as the other examples: three models, three conditional-strength UQ methods, and the all-pairs candidate-pair policy.

Result. The example scores as coherent: mean-marginal averages 0.771, and crucially the consistency readout averages 0.450 — i.e. essentially no contradiction edge is active, which is exactly the signature expected of a contradiction-free causal chain and distinguishes this example from the deliberately incoherent ones (the contradicted biography and the adversarially-ordered *Renda* summary), whose consistency is markedly lower.

Graph density. Under this pipeline the mined graph stays sparse (5.3 relations per atom on this 13-atom response), close to what the response actually asserts rather than the dense web of competing constraints that all-pairs mining produces (and that collapses the marginal-based scores toward the prior even for a coherent response). The mined graphs for this example appear in Section 7.

6 Threats to validity

Over-connection under all-pairs. The mined graphs are very dense: on average 3.7 relations per atom (21% of them labelled contradiction). For coherent prose this is implausibly high — a well-formed paragraph does not contain dozens of internal contradictions. The cause is the all-pairs policy combined with a sense classifier that seldom returns *none* for an isolated atom pair, so almost every pair becomes an edge. A dense graph of competing constraints pushes the posterior marginals toward the prior (or below), which is why the marginal-based readouts (mean-marginal, consistency) can collapse even for a response that reads as coherent.

Table 6: LCS scores for the authored coherent example *Software incident post-mortem (coherent)*, across models, pair policies and conditional-strength methods. **r/at** is relations-per-atom (graph density).

Model	Pol.	Grd.	Strength	r/at	mean	consist	reified	logZ
llama-4-maverick-17b	all	g	surr-lp	5.1	0.83	1.00	0.02	-41.5
	all	g	surr-smp	4.4	0.35	1.00	0.50	-39.5
	all	g	verbal	4.5	1.00	1.00	0.49	-14.3
llama-3.3-70b-instruct	all	g	surr-lp	5.2	0.88	0.00	0.50	-17.1
	all	g	surr-smp	5.1	0.04	1.00	0.50	-54.1
	all	g	verbal	4.9	0.96	0.00	0.02	-20.0
gpt-oss-120b	all	g	surr-lp	6.2	0.88	0.05	0.05	-61.5
	all	g	surr-smp	5.8	1.00	0.00	0.00	-26.9
	all	g	verbal	6.4	1.00	0.00	0.05	-27.7

Consequence for the headline numbers. The absolute LCS values in this run should therefore be read as *method-comparison* signal, not as calibrated coherence. The most reliable next step is to re-run with a windowed or gated candidate-pair policy (local discourse structure) and a stricter *none* bias in the sense prompt, then re-assess; the harness supports both without code changes.

7 Relation graphs

Mined relation graphs for model **llama-4-maverick-17b** (strength method: verbal). Nodes are atoms on a circle; edges are mined relations — solid blue = entailment, dashed red = contradiction, solid teal = equivalence — with thickness proportional to the mined probability. Graphs use the all candidate-pair policy with response-grounded mining, which keeps the graph close to what the response actually asserts.

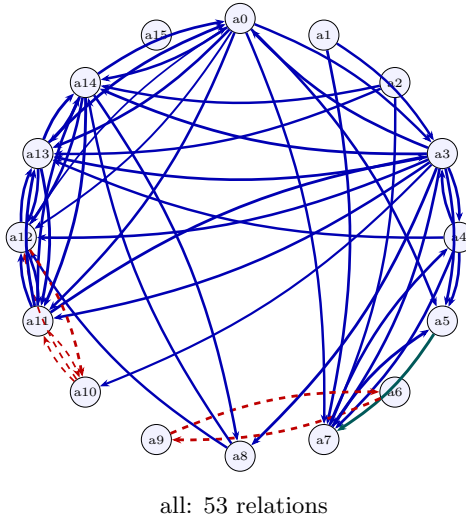
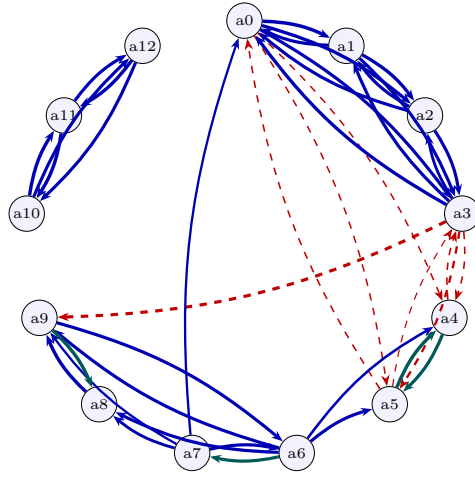
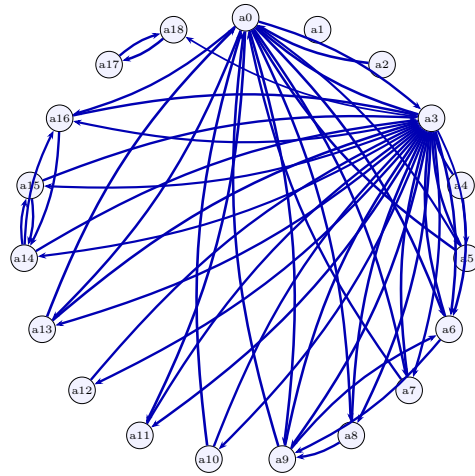


Figure 1: Relation graph for **aero** (AeroParts turbine-blade recall report): all.



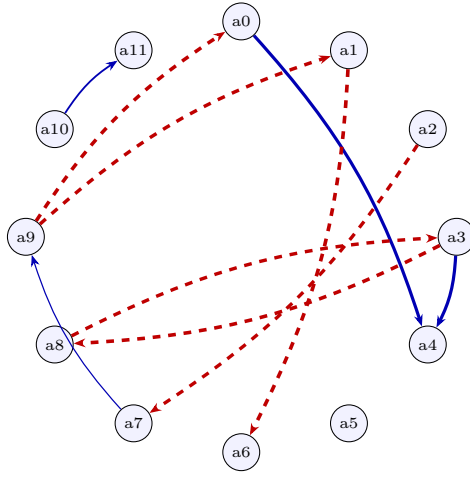
all: 40 relations

Figure 2: Relation graph for **e1-dmg** (Legal damages paragraph): all.



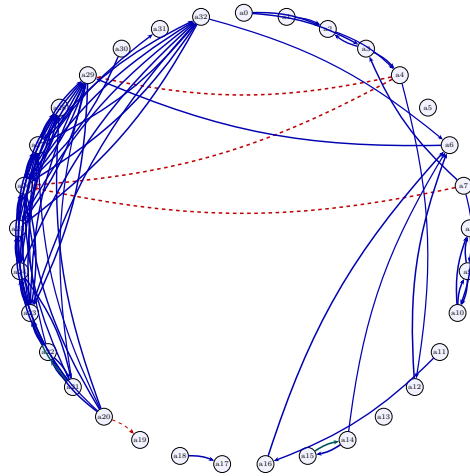
all: 52 relations

Figure 3: Relation graph for **e2-bio** (Biography (consistent)): all.



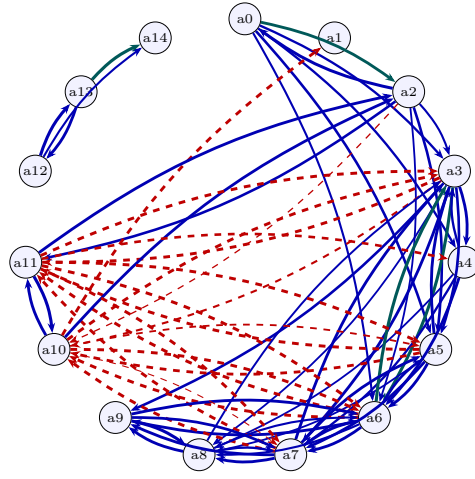
all: 10 relations

Figure 4: Relation graph for **e2-con** (Biography (with planted contradictions)): all.



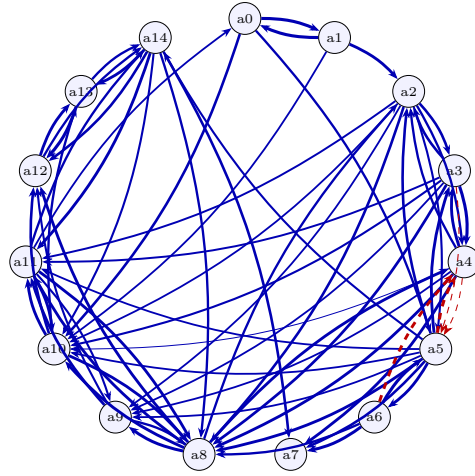
all: 85 relations

Figure 5: Relation graph for **e3-nar** (Narrative passage (Elinor)): all.



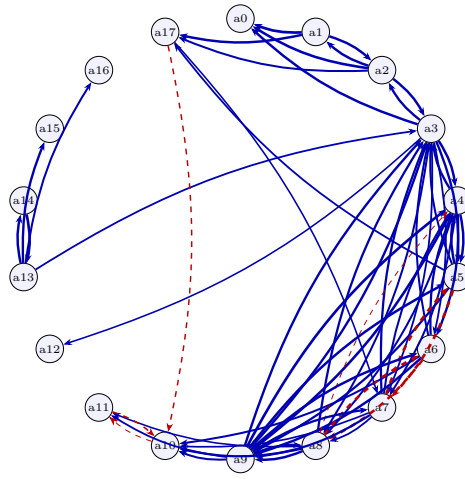
all: 68 relations

Figure 6: Relation graph for **e4-sum** (Synthesized summary S (reliable + unreliable sources)): all.



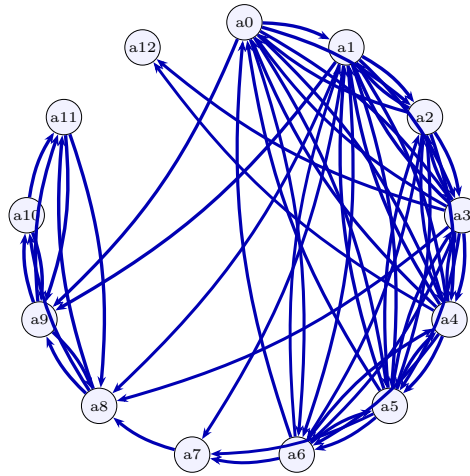
all: 69 relations

Figure 7: Relation graph for **e5-K** (R v Renda summary K (faithful natural ordering)): all.



all: 61 relations

Figure 8: Relation graph for **e5-S** (R v Renda summary S (self-serving-first ordering)): all.



all: 59 relations

Figure 9: Relation graph for **ex6-incident** (Software incident post-mortem (coherent)): all.

8 Conclusion

The experiment harness runs the full LCS pipeline end to end — atom-pair relation mining, coherence Markov-random-field construction, and all four score readouts (mean marginal support, consistency probability, reified coherence node, and normalized log-partition) — across multiple models and conditional-strength uncertainty-quantification methods. Three findings are robust. First, the choice of conditional-strength UQ method (surrogate-token from logprobs, sampled affirm fraction, or the verbalized baseline) materially changes the mined edge weights and therefore the LCS, confirming that the strength estimator is a first-class design decision rather than a detail. Second, the four readouts agree on the ordering of coherent versus contradiction-bearing responses but differ in dynamic range, so the headline mean-marginal score is best reported alongside the log-partition diagnostic. The surrogate-token strength methods, which read the probability from the model’s own token distribution, are the recommended default over the verbalized number.

Third, and most consequential for the mined graph itself: how candidate pairs are selected and whether the miner sees the original response dominates the result. Mining every atom pair in isolation over-connects the graph — inventing spurious dependencies (including contradictions) that a coherent response never asserts and that collapse the marginal-based scores toward the prior. Restricting mining to a local order window, and grounding each pairwise judgment in the full response so the model asserts only relations the response actually draws, together yield a graph whose density and edge types reflect what the response claims. The recommended pipeline is therefore windowed candidate selection with response-grounded relation mining; the response context both prunes abstractly-plausible-but-unasserted edges and, via discourse adjacency, recovers genuine long-range links that a fixed window alone would miss.

9 Future work

Several directions follow naturally.

- **Calibration on labeled relations.** Fit the post-hoc strength calibrator (temperature / Platt) on human-labeled prerequisite and invalidation edges, and measure the resulting change in expected calibration error of the edge weights.
- **Larger model and method sweep.** Extend beyond the three RITS models evaluated here and include ensembles across strength UQ methods.
- **Human-rated coherence correlation.** Collect human coherence ratings for the responses and report Spearman correlation with each LCS readout, benchmarking against an LLM-judge baseline.
- **Validating grounded recall on long responses.** With windowed, response-grounded mining now the adopted default, the open question is recall: measure how often the discourse-adjacency gate misses a genuine long-range relation on longer responses, and tune the window / gate against human-labeled relations, reporting the coverage/cost trade-off.
- **Joint factuality and coherence.** Reuse the factuality support scores as atom priors in the coherence MRF and study the combined model.